AN EXACT FOURIER BOUND WITH EXPONENT SAVING 63/50000

Abstract. An explicit finite tensor construction gives a deterministic exact discrete Fourier transform at every positive integer length n, with cost $O(n(\log n)^{1-63/50000})$. The model permits unrestricted complex coefficients and one specified root of polynomial order, and charges scalar preparation, schedule construction, indexing and movement. The construction uses a 73-axis closure of the public C588AT provider, with arbitrary live helpers and separately charged initialized copies. Its exact recursive moment contracts at tensor saving $1261/10^6$; strict exponent slack absorbs the all-length Fourier transfer loss. The proof combines an explicit finite certificate with the charged all-length transfer.

1. Statement, model and provenance

For $n \geq 1$, let $\zeta_n = \exp(2\pi i/n)$ and $F_n = (\zeta_n^{jk})_{0\leq j,k<n}$. The model is that of OpenAI's all-length exact Fourier theorem [1]. Complex field operations have unit cost. Operations on variable data are linear: additions, subtractions and multiplication by prepared input-independent scalars. Scalar preparation may divide only by proved nonzero quantities. Integer operations and random access on a fixed number of $O(\log(n+2))$-bit words have unit cost. Preparation, initialization, table generation, indexing and movement are included. Coefficients and intermediate magnitudes are unrestricted. One explicitly specified root of unity is supplied at unit cost; other roots are prepared from it.

Theorem 1.1. *Put $\delta = 63/50000$. A single deterministic algorithm computes $F_n x$ exactly for every positive integer n and every $x \in \mathbb{C}^n$, with charged cost*

$$O\big(n(\log n)^{1-\delta}\big) = O\big(n(\log n)^{49937/50000}\big) \qquad (n \longrightarrow \infty).$$

It needs only the specified root $\zeta_{D_*(n)}$, where $D_*(n)$ is the explicit integer in (10) *and $D_*(n) < 1024n^3$. Unrolling the dimension-dependent schedule gives exact complex linear circuits of the same asymptotic size. In particular, $\delta > 0.00125$.*

The proof has four interfaces: a rational provider with verified scalar and frame identities, a simultaneous tensor program on every live role, a charged recurrence allowing separate initialized workspace, and a charged transfer to all Fourier lengths. A finite call histogram by itself proves none of these interfaces. The complete expanded ordinary arguments are retained in the accompanying `proof.md`; the full finite certificate and three standalone exact checkers are also supplied. The exposition below follows those arguments without changing their scope.

The all-length reduction and exact-width compiler come from OpenAI [1, 2]. The second cited paper's nonuniform subsequence result has different quantifiers and is not substituted for the all-length theorem. The public complex588 package and its batching framework are attributed to their source contributors [3]. The C588AT data, generalized frames and compressed closure are taken from the immutable verifier package [4]; the proof supplies their actual complex-array semantics and physical accounting.

The exponent saving improves OpenAI's all-length bound from 10^{-13} to $63/50000 = 0.00126$ in the same exact-arithmetic model. The complete Lean development accompanies this paper and exports `Race.bound`. It uses Lean 4.34.1 and a pinned mathlib version.

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2. Explicit subspace frames

Write $\alpha_0 = (1+i)/2$, $\beta_0 = (1-i)/2$ and

$$C = \begin{pmatrix} \alpha_0 & \beta_0 \\ \beta_0 & \alpha_0 \end{pmatrix}, \qquad C^2 = X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad C^{-1} = XC.$$

For $V = \mathbb{F}_2^d$, put $D(x) = i^{|x|}$, where $|x|$ is Hamming weight. Let P_S apply C on the binary coordinates in S and identity elsewhere. For an invertible binary coordinate map G, define

$$K_G(x,y) = \mathbf{1}_{Gx=y}\frac{D(x)}{D(y)}, \quad U(G,S) = G^{-1}\mathbb{F}_2^S, \quad T(G,S) = K_G P_S K_G^{-1}.$$

A frame is a matrix invariant: an array carrying frame T already equals Tz for its abstract value z. Maintaining this invariant does not instruct the algorithm to evaluate a dense T or its inverse.

Lemma 2.1 (Charged frame transitions)**.** *Frames representing the same binary subspace differ by an explicitly constructible invertible monomial left adapter over $\mathbb{Q}(i)$. Between nested represented subspaces with dimension difference $0 < r < d$, either direction uses exactly one adapted forward $C^{\otimes r}$ block, besides monomial work. Under f-column replication and b spectator bits, it is one adapted batch of $C^{\otimes rf}$, and all remaining work, preparation and storage are $O(2^{df+b})$, with $O(df+b+1)$-bit addresses. Each complementary fiber of the batch must be charged separately.*

Proof. The identity $D(x+y) = D(x)D(y)(-1)^{x\cdot y}$ gives

$$T(G,S)(x,y) = \mathbf{1}_{x+y\in U(G,S)}\alpha_0^{|S|}\frac{D(x)}{D(y)}(-1)^{(Gx)\cdot G(x+y)}. \tag{1}$$

Suppose $U = U(G,S) = U(H,R)$ and put $v_j = G^{-1}e_j$ for $j \in S$. Define

$$p(x) = \sum_{j\in S}\big((Gx)_j + (Hx)\cdot Hv_j\big)v_j, \qquad Bx = x + p(x).$$

Then $p(x) \in U$ and $(GBx)\cdot Gu = (Hx)\cdot Hu$ for every $u \in U$. If $Bx = 0$, then $x \in U$ and Hx is orthogonal to all of the coordinate space $H(U) = \mathbb{F}_2^R$, so $x = 0$. Hence B is invertible. The monomial

$$A(x,z) = \mathbf{1}_{Bx=z}\frac{D(x)}{D(Bx)}(-1)^{(GBx)\cdot Gp(x)}$$

satisfies $T(H,R) = AT(G,S)$ by substitution in (1).

For nested subspaces choose, by binary elimination, one coordinate map B_0 representing both with nested coordinate sets $I \subset J$. Change representatives using the adapters just constructed, and use

$$T(B_0,J) = K_{B_0}P_{J\setminus I}K_{B_0}^{-1}T(B_0,I).$$

The reverse move uses $P_{J\setminus I}^{-1} = X_{J\setminus I}P_{J\setminus I}$. Thus only one forward block is paid in either direction.

All finite binary tables and phases are generated and charged. Tensoring across f columns collects the selected rf bits by a coordinate permutation. Enumerate column digits by depth-first traversal of the complete radix-2^d prefix tree, carrying the image address and phase product. The tree has $O(2^{df+b})$ nodes including spectators; each edge costs bounded word work. The same traversal packs selected and unselected bits. Two array buffers and a prefix stack give the asserted space and address bounds, without an extra factor f per entry. □

3. The actual C588AT invocation

The supplied certificate `code/c588at-provider.json` has 241357 bytes and SHA-256

`b52b8e31dd731c7bdc1fa3e556224ffa51c76dc66ca36849ecec8c85733f9624.`

It is a declarative extraction of the pinned public C588AT data, not an assumed source theorem. It contains 112 port vectors and orthobases, 588 helper entrance labels, eight retained centers, 1082 RREF frames, and every rational gate and coefficient [4].

In an admissible window $V = A \oplus E(\mathbb{F}_2^9) \oplus D_0$, choose the ordered basis with A first and the window next. An RREF label U with pivot rows b_p has involution $G_U x = x + \sum_p x_p(b_p + e_p)$, taking U to its pivot coordinates. This and the chosen window basis define a frame $\Phi(U)$ representing $A \oplus E(U)$. The helper's entrance frame is $E_q = \Phi(U_q)$; its full-window frame F_q uses the same coordinate map and all window coordinates.

Proposition 3.1. *The certificate realizes both scalar maps*

$$(X, Y, S) \mapsto (X, Y + X, S), \qquad (X, Y, S) \mapsto (X - Y, Y, S)$$

on arbitrary live values. Port t starts in frames $\Phi(\operatorname{span} u_t), \Phi(0)$ and ends in frames $\Phi(\mathbb{F}_2^9), \Phi(u_t^\perp)$. A helper arriving in any frame Z_q leaves in $F_q E_q^{-1} Z_q$, preserving its scalar value. Both maps use eight separately initialized copies and the same recursive profile

$$(h_1, h_2, h_3, h_4, h_5, h_6, h_8) = (2568, 492, 280, 232, 56, 56, 92).$$

The helper residual ranks $9 - \dim U_q$ have multiplicities $\{9 : 360, 8 : 140, 7 : 32, 5 : 8, 4 : 48\}$. The statement holds under every column replication, with charged linear remaining work as in Lemma 2.1.

Proof. Each listed gate visits a frame before its scalar additions. The checker verifies strict nested visits, equal labels at every addition, all orthobases and endpoints, and the entrance-gauged set G of 228 helpers. The two rational phase matrices are A_s, B_s. Let C_s select the eight retained centers and J_s be the listed scatter matrix. With $J(X, Y, S) = (X, Y + J_s C_s S, S)$, exact backward covector propagation verifies

$$M = B_s J A_s = \begin{pmatrix} I & 0 & 0 \\ I & I & K \\ L & 0 & H \end{pmatrix}, \qquad K_{[:,G]} = 0. \tag{2}$$

The untested initial-Y zero blocks follow from the explicitly checked triangular gate supports. Every factor of M is invertible; if $Hs = 0$, then $M(0, -Ks, s) = 0$, so $s = 0$ and H is invertible. Prepare K, L, H^{-1} by finite exact rational operations. The corrections

$$g_1(X, Y, S) = (X, Y - KS, S), \quad g_8(X, Y, S) = (X, Y, H^{-1}(S - LX))$$

give $g_8 M g_1(X, Y, S) = (X, Y + X, S)$. The first correction uses only ungauged helpers at the same frame as Y; the last mixes only full-window X, S arrays, so both are legal pointwise operations.

The scatter cannot directly mix unequal frames. Initialize eight new arrays to zero, copy the retained helper values at their center frames, move these copies down to the zero label by eight rank-eight blocks, and scatter into Y. Erase the copies afterward with charged writes. No live helper is assumed zero. The phase visits plus these eight blocks give precisely the stated profile: phase A has ranks $\{1 : 1028, 2 : 8, 4 : 56\}$, phase B has $\{1 : 1540, 2 : 484, 3 : 280, 4 : 176, 5 : 56, 6 : 56, 8 : 84\}$.

Replace each rational shear $I + ce_t e_s^T$ by $I - ce_s e_t^T$ in the same chronological order, retaining its frame visits. This computes the inverse transpose of the scalar product, since $(DE)^{-T} = D^{-T} E^{-T}$. The corrections remain legal. Realize J^{-T} by copying $J_s^T Y$ at zero frame into the extra arrays, moving them up to their centers, and adding their negative selected values to the helpers. The resulting scalar map is $(g_8 M g_1)^{-T}$, namely $(X - Y, Y, S)$, at the same price.

Finally change the helper's full-window representative to F_q by a monomial. For arbitrary incoming $Z_q s$, interpret its initial scalar as $E_q^{-1} Z_q s$; this is semantic interpretation, not a dense inverse operation. Scalar restoration therefore gives $F_q E_q^{-1} Z_q s$. The product $F_q E_q^{-1}$ is exactly a monomial-conjugated P_R on the nonpivot window coordinates, of rank $9 - \dim U_q$.

For orthogonal G, the ratio $D(Gx)/D(x)$ is a sign character. Consequently $T(G, S)$ differs from the ordinary product of C along the orthonormal directions $G^{-1} e_j$ by a monomial: each negative

factor uses $\alpha_0 I - \beta_0 X_z = iX_z(\alpha_0 I + \beta_0 X_z)$. Thus the supplied orthobases give the needed boundary representatives. All identities tensor across columns. Finite rational corrections, copy writes and the adapters of Lemma 2.1 are charged linear overhead. The full gate definitions and exhaustive finite verification are supplied with the expanded proof. □

4. The 73-axis closure on every live array

Let $V = H \oplus E_0 \oplus \cdots \oplus E_7$, where H has coordinates $0, \dots, 8$ and each E_c has eight coordinates. Put $p = 0$, $H' = H \setminus \{p\}$ and $R_{<k} = \bigcup_{c<k} E_c$. Define the finite nonempty class set

$$\Gamma = \{g \in \mathbb{F}_2^{73\times 73} : g^T g = I\}, \qquad Q = |\Gamma|.$$

Lexicographic enumeration of all binary matrices defines the set and all its tables constructively. This enormous finite preparation is charged; Q is fixed independently of the eventual input length.

For port t, let B_t be its supplied orthobase, $D_t = \operatorname{diag}(B_t, I_{64})$, and $\mathcal{F}_{g,t}(S) = T((gD_t)^{-1}, S)$. For each (g,t) use eight replicas, each containing four pairs (X_j, Y_j), $1 \le j \le 4$. Initially X_j has frame $\mathcal{F}(\{p\})$ and Y_j frame I. These are invariants for arbitrary physical inputs. With 4824 live helper banks per class, the live count and separate copy count are

$$W = Q(8 \cdot 4 \cdot 2 \cdot 112 + 4824) = Q \cdot 11992, \qquad \text{initialized copies} = 8Q. \tag{3}$$

Serial movement buffers are additional charged storage.

At stage $k = 0, \dots, 8$, invocation class d uses background $d \operatorname{span} R_{<k}$ and window dH. Stage zero serves class g with $d = g$. For $k \ge 1$, let Π_{k-1} swap H' with E_{k-1}, fixing p and other coordinates, and set

$$R_{t,k} = D_t \Pi_{k-1} D_t^{-1}, \qquad d = gR_{t,k}.$$

Each $R_{t,k}$ is an orthogonal involution, so this is a permutation of Γ for each port. Every data role is used once at its stage. In the class basis, $(gD_t)^{-1}d = \Pi_{k-1} D_t^{-1}$. Since $D_t^{-1} u_t = e_p$ and $D_t^{-1}(u_t^{\perp}) = \operatorname{span} H'$, the stage-$k$ output frames represent

$$X[k] = H \cup R_{<k}, \qquad Y[k] = H' \cup R_{<k}.$$

For $k \ge 1$ the inputs are exactly $X[k-1], Y[k-1]$; at stage zero they are $\{p\}, \varnothing$. Same-subspace adapters select the common $\mathcal{F}_{g,t}$ representatives. These generator identities hold for all g by left multiplication, not by an experimental enumeration.

4.1. **A run on four pairs.** Run stage zero forward on pair 1, then stage 1 backward on pair 1. For $j = 2, 3, 4$ perform the following fixed sequence. Move the idle pair j to $X[2j-3], Y[2j-3]$, paying rank $16(j-1)$ on each side; do $X_{j-1} \mathrel{+}= X_j$, $Y_j \mathrel{-}= Y_{j-1}$; run stage $2j-2$ forward on pair $j-1$; move pair j up by rank eight on each side; do $Y_j \mathrel{+}= Y_{j-1}$, $X_{j-1} \mathrel{-}= X_j$; and run stage $2j-1$ backward on pair j. Run stage 8 forward on pair 4. Finish pairs 1, 2, 3 at the common endpoints $V, V \setminus \{p\}$ by rank 48, 32, 16 moves on each side. Finally, for $j = 4, 3, 2$, do $Y_{j-1} \mathrel{-}= Y_j$, $X_j \mathrel{+}= X_{j-1}$. Every scalar addition connects equal frames. Before those final gates, the scalar values are

j	X_j	Y_j
1	$-y_1$	$x_1 + x_2$
2	$y_1 - y_2$	$x_2 + x_3$
3	$-y_1 + y_2 - y_3$	$x_3 + x_4$
4	$y_1 - y_2 + y_3 - y_4$	x_4

so the final gates give $(x_j, y_j) \mapsto (-y_j, x_j)$. The extra price per port across the eight replicas is

$$48\phi(8) + 32\phi(16) + 32\phi(32) + 32\phi(48), \tag{4}$$

where $\phi(r)$ denotes one rank-r adapted batch.

For the final correction write $\mathcal{F}(S) = KP_S K^{-1}$ and $L_0 = \{p\}$. The full-subspace adapter gives a monomial N with $N\mathcal{F}(V) = P_V$. Also $\mathcal{F}(V)\mathcal{F}(L_0) = J\mathcal{F}(V \setminus L_0)$, where $J = KX_pK^{-1}$ is monomial.

Apply N to each X, NJ to each Y, then replace the physical pair (X, Y) by $(Y, -X)$. The outputs are $P_V\mathcal{F}(L_0)x_j, P_V y_j$, the full tensor applied to the original physical inputs. This avoids an initial dense frame inverse.

4.2. **Hosting residuals on arbitrary helper banks.** Maintain each bank in frame AP_U with A monomial. Suppose its next already-paid invocation residual is $C_0P_RC_1$, with $|R| = r$ and $|U| + r \le 73$. Choose the first r unused coordinates T and a coordinate permutation σ taking T to R. Its address map M_σ satisfies $P_RM_\sigma = M_\sigma P_T$. Apply the monomial $C_1^{-1}M_\sigma A^{-1}$ before the invocation. The bank then has frame

$$(C_0P_RC_1)(C_1^{-1}M_\sigma A^{-1})(AP_U) = (C_0M_\sigma)P_{T\cup U}. \tag{5}$$

Only monomials are inverted. At the end apply A^{-1} and, if needed, one block on the unused coordinates. The bank's arbitrary scalar input is preserved throughout, and its final physical map is P_V.

Partition the 588 residual slots into the following 67 groups. Within each weight use increasing slot order.

Groups	Weight 9	8	7	5	4	Load
30	6	1	1	0	1	73
18	5	3	0	0	1	73
3	7	0	0	2	0	73
1	6	0	2	1	0	73
7	1	8	0	0	0	73
7	8	0	0	0	0	72
1	0	0	0	1	0	5

At substage $k = 0, \dots, 71$, assign position j of group a to bank $72a + ((k + j) \bmod 72)$. This is injective at a substage, and every bank gets every slot of its group once over 72 substages. Positive partial loads never exceed final loads. Thus (5) is always legal. The final padding multiset is $\{0 : 4248, 1 : 504, 68 : 72\}$.

Proposition 4.1. *For every $f \ge 1$ and $0 \le b < 73$, the construction applies $I_{2^b} \otimes C^{\otimes 73f}$ to all $W = Q \cdot 11992$ arbitrary live arrays of length 2^{73f+b}. Its recursive counts are Qn_r, where n_r is in Table 1. Besides the $8Q$ initialized copies, it uses fixed serial buffers. All nonrecursive work and storage are $O((W + S)2^{73f+b})$ for a fixed finite S, including table generation, copy initialization, scalar corrections, adapters, movement and indexing. Addresses have $O(73f + b + 1)$ bits.*

Proof. There are 72 invocations per class. Their profiles, the data climbs in (4), and the padding give $n_1 = 72 \cdot 2568 + 504$, $n_8 = 72 \cdot 92 + 112 \cdot 48$, $n_{16} = n_{32} = n_{48} = 112 \cdot 32$, $n_{68} = 72$, and the remaining entries $72h_r$. Each rank is below 73; their weighted sum is $872856 = 73 \cdot 11992 - 2560$. All data and banks have the full tensor as proved above. Multiplying role and call counts by Q retains the actual physical construction. Every table has a fixed terminating generation algorithm. Column-prefix traversal from Lemma 2.1 implements all replicated monomials in linear work. Reuse the eight copies per class between substages; initialize and erase them with charged writes. Fixed correction and movement buffers are separately counted. No live input is replaced by zero. □

5. Exact contraction and charged amplification

Put $a = 1261/10^6$ and $\alpha = 1 - a$. Table 1 supplies both the exact call counts and rational logarithm bounds used below. For $1 \le t \le 2$, $z = (t - 1)/(t + 1)$ satisfies

$$L(t) = 2\sum_{j=0}^{39} \frac{z^{2j+1}}{2j+1} \le \log t \le H(t) = L(t) + \frac{2z^{81}}{81(1-z^2)}.$$

r	n_r	u_r with $\log(73/r) \le u_r/10^9$
1	185400	4290459442
2	35424	3597312261
3	20160	3191847153
4	16704	2904165081
5	4032	2681021529
6	4032	2498699972
8	12000	2211017900
16	3584	1517870719
32	3584	824723539
48	3584	419258431
68	72	70951736

TABLE 1. Exact profile per orthogonal class and rational rate certificate.

This is the positive integrated geometric series and its tail bound. Write $73/r = 2^s t$ with $1 \le t < 2$; exact rational arithmetic checks $sH(2) + H(t) \le u_r/10^9$. Since $j! \ge 2 \cdot 3^{j-2}$ for $j \ge 2$,

$$e^x \le E(x) := 1 + x + \frac{x^2}{2(1 - x/3)} \quad (0 \le x < 3).$$

The supplied rate checker verifies the finite rational comparison

$$\Psi = \frac{1}{11992} \sum_r n_r (r/73)^{1-a} \le \sum_r \frac{r n_r}{73 \cdot 11992} E(a u_r/10^9) < \frac{999989}{10^6} < 1. \tag{6}$$

For the actual program, $(Q \cdot 11992)^{-1} \sum_r Q n_r (r/73)^\alpha$ equals Ψ exactly. This cancels a common multiplicity, not physical roles or workspace.

For completeness, let $T(k)$ be simultaneous cost on W arbitrary arrays and $t(k) = T(k)/(W2^k)$. For $k \ge 73$, put $f = \lfloor k/73 \rfloor$, $b = k - 73f$. Apply the b spectator factors directly, then the concrete program of Proposition 4.1. A rank-r call has 2^{k-rf} fibers. Recurse on each complete group of W fibers; use the elementary tensor algorithm on the fewer than W leftovers. Allocate each child's initialized workspace separately. With $N_r = Q n_r$, the charged recurrence is

$$T(k) \le A_0(W + S)2^k + \sum_r N_r \left[\left\lfloor \frac{2^{k-rf}}{W} \right\rfloor T(rf) + B_0(W - 1) rf\, 2^{rf} \right].$$

Because $rf\, 2^{rf-k} \le rf 2^{-f} \le r$, normalization gives $t(k) \le A_1 + W^{-1} \sum_r N_r t(r\lfloor k/73 \rfloor)$ with fixed A_1, including scratch and leftovers. For positive k, strong induction on k^α, using $rf \le rk/73$, yields $t(k) \le K k^\alpha$ once K covers $1 \le k < 73$ and $K 73^\alpha (1 - \Psi) \ge A_1$. Dimension zero is identity. This does not use an inequality with $rf + 1$ in place of rf.

For a single input initialize $W - 1$ additional arrays and keep its output. Thus

$$C^{\otimes k} \text{ costs } O(2^k (k+1)^{1-a}), \qquad k \ge 0. \tag{7}$$

Serial children have strictly smaller integer dimensions, so their array lengths sum along a stack to less than 2^{k+1}. Memory is $O(2^k)$ and addresses have $O(k + 1)$ bits. Every child's arbitrary input may be a parent's nonzero scratch value; none is presumed zero. The incomplete-batch mechanism is also present in the pinned public framework [3]; the expanded proof states and proves the version with separate charged initialized workspace.

6. Charged transfer to every Fourier length

This section records the complete transfer interfaces of the accepted proof, following OpenAI's exact-width method [1]. Part E of `proof.md` gives the expanded compiler and address proof, including all finite fallbacks. Its tensor hypothesis is now supplied by (7).

6.1. **A local compiler without extra coordinates.** For primitive rth root u, let column k of P be the coefficient vector of $\prod_{j<k}(Z-u^j)$. Then $N=F_r(u)P$ is lower triangular. With $H_0=1$, $H_j=\prod_{s=1}^{j}(1-u^s)\neq 0$ for $0<j<r$,

$$N_{ik}=(-1)^k u^{k(k-1)/2}\frac{H_i}{H_{i-k}}\quad (i\geq k).$$

Thus N factors as a diagonal matrix, a lower Toeplitz matrix, and a diagonal matrix. If $D_N=\operatorname{diag}(N_{kk})$, uniqueness of unit-lower/diagonal/unit-upper factorization and symmetry of F_r give

$$F_r(u)=ND_N^{-1}N^T. \tag{8}$$

A fixed-vector convolution has a printed bounded-fan-in/fan-out linear DAG with $O(v\log(2+v))$ gates and $O(\log(2+v))$ depth, by the binary FFT. Such a DAG for $M:\mathbb{C}^e\to\mathbb{C}^b$, using g gate coordinates, can be replayed on arbitrary borrowed values to implement $(x,y,z)\mapsto(x,y+Mx,z)$. Indeed a forward additive sweep has form $z\mapsto Uz+Tx$; if the output reads are $Fz+Dx$, then $FT+D=M$. Sweep, add output, and undo; then sweep with all direct x edges omitted, subtract its output, and undo. The added FUz terms cancel and z is restored. This uses at most $8g+2b$ shears. Edge coloring at each depth level gives $O(H+1)$ disjoint-pair layers for bounded fan-out. The identity allows borrowed values correlated with all other coordinates.

To compile a lower triangular Toeplitz matrix T_v with nonzero diagonal h_0, prepare its inverse sequence $g_0=h_0^{-1}$, $g_k=-h_0^{-1}\sum_{j=1}^{k}h_jg_{k-j}$. For $p=\lfloor v/2\rfloor$, factor

$$T_v=\begin{pmatrix}I & 0\\ E & I\end{pmatrix}\operatorname{diag}(T_p,T_{v-p}),\qquad E=BT_p^{-1}.$$

Its entries satisfy $E_{ij}-E_{i-1,j-1}=-h_{i-p}g_{p-j}$ away from chunk boundaries. Therefore each $b\times e$ chunk M has displacement $M-J_bMJ_e^T$ equal to three explicit outer products, including the first row/column corrections. The identity $M=\sum_{\ell\geq 0}J_b^{\ell}(M-J_bMJ_e^T)(J_e^T)^{\ell}$ evaluates a chunk with a fixed number of convolutions. Print these graphs and choose the largest chunk size for which every graph fits with its source and target in the same v coordinates. A size $\Theta(v/\log v)$ fits for all sufficiently large v; otherwise use the finite direct fallback. The $O(\log^2 v)$ chunk pairs each take $O(\log v)$ replay layers. Balanced child recursion gives $O(1+\log^4 r)$ layers on exactly r coordinates. All graph printing, fit tests, coefficient generation and edge coloring are polynomial in r, with an absolute degree. Borrowed coordinates are restored after each chunk. Equation (8) and transposition preserve the layer types.

Every scalar shear can use one common pattern of six forward C slots. To see this, put $S_0=\operatorname{diag}(1,i)$ and $H_2=\left(\begin{smallmatrix}1 & 1\\ 1 & -1\end{smallmatrix}\right)$. Direct multiplication gives $H_2=\alpha_0^{-1}S_0CS_0$ and

$$H_2\operatorname{diag}(2,1)H_2\operatorname{diag}(-3,1)H_2=K_0=\begin{pmatrix}-8 & -10\\ 0 & -6\end{pmatrix}.$$

For $t\neq 0$, conjugating $\operatorname{diag}(-1/8,-1/6)K_0$ by $D_t=\operatorname{diag}(4t/5,1)$ gives the upper shear of coefficient t. For an arbitrary prepared μ, set $\kappa=1+\mu\overline{\mu}$. Both $t_1=\kappa$ and $t_2=\mu-\kappa$ are nonzero, since $1+|\mu|^2>|\mu|$. Their shears compose to the μ shear. Conjugates are prepared by evaluating the coefficient DAG at inverse roots and $-i$, not by conjugating variable data or testing complex zeros. This proves the common slot pattern even for zero coefficients. All needed extra roots have power-of-two order less than $8r$.

6.2. **Synchronization and linear indexing.** For widths $r_1, \dots, r_\ell \geq 2$ and $R = \prod_j r_j$, the mixed-radix prefix tree has fewer than $2R$ nodes. Tensor monomials are implemented by carrying original and permuted address prefixes and a running coefficient. The cost is $O(R)$, rather than $O(R\ell)$.

At a simultaneous C slot each local axis splits into ordered pairs and singletons. A sector chooses one block on each axis, with sizes $q_j \in \{1, 2\}$; its map is $C^{\otimes k}$ and width $2^k = \prod_j q_j$. If p_j is the number of local coordinates preceding its block and $\tau_j = \prod_{i>j} r_i$, its packed start is

$$S = \sum_{j=1}^{\ell} \left(\prod_{i<j} q_i \right) p_j \tau_j.$$

This classifies earlier sectors by their first different axis. During a digit traversal with local coordinate a_j and within-block digit t_j, update

$$S \leftarrow S + Q_0 p_j \tau_j, \quad Q_0 \leftarrow Q_0 q_j, \quad o \leftarrow o q_j + t_j, \quad I \leftarrow I r_j + a_j.$$

The packed address is $S + o$ and the original address I. Packing, unpacking and sector listing therefore cost $O(R)$ in total. The sector widths sum to R and $k \leq \ell$; (7) bounds all sector transforms by $O(R(\ell+1)^\alpha)$. Padding local compiler words to common length gives

$$O\big(R(\ell+1)^\alpha (1 + \log r_{\max})^4\big) + \mathrm{poly}(\ell, r_{\max}, \log(R+2)). \tag{9}$$

The polynomial is charged preparation with absolute degree. Local words are matrix identities on arbitrary states; tensor-product composition remains valid when other axes alter their borrowed values.

6.3. **Working lengths, CRT and the sole supplied root.** Let r_j be the odd primes in order and choose maximal $\ell \geq 0$ with $R = \prod_{j=1}^{\ell} r_j \leq 2n$. Choose least $e \geq 0$ with $L = R2^e \geq 2n$. Then $2n \leq L < 4n$ and $2^e < 2r_{\ell+1}$. Elementary prime estimates suffice: the central binomial coefficient satisfies

$$\frac{4^u}{2u+1} \leq \binom{2u}{u} \leq (2u)^{\pi(2u)}.$$

Each prime contributes at most $2u$ to the latter coefficient. With $u = (j+1)^2$ this gives $r_j = O((j+1)^2)$. Since $r_j \geq j+2$, $\log R = \Theta(\ell \log(\ell+2))$. Maximality implies

$$\ell = \Theta(\log n / \log\log n), \qquad \log \ell = \Theta(\log\log n), \qquad e = O(\log(\ell+2)).$$

Trial division finds the needed primes with polynomial logarithmic work. Initial lengths are handled directly.

For coprime factors $q \mid L$, put $E_q = (L/q)d_q$, where $(L/q)d_q \equiv 1 \pmod q$. The CRT idempotents give

$$j = \sum_q E_q j_q \pmod L, \qquad \zeta_L^{jk} = \prod_q \zeta_q^{d_q j_q k_q}.$$

Prefix traversal with tabulated residues implements the CRT and local output permutations in $O(L)$ work. Apply (9) to odd-prime axes, and the elementary binary FFT to the 2^e axis. All tables, twiddles and compilation are charged, giving

$$T(F_L) = O\big(L[(\ell+1)^\alpha (1 + \log(\ell+2))^4 + 1 + e]\big) + \mathrm{poly}(\log(n+2)).$$

The inverse has the same cost since $F_L^2 = LJ$, with J negating the index. Its only additional division is by the nonzero integer L.

For arbitrary n, put $\eta = \zeta_{2n}$. The chirp identity

$$\zeta_n^{jk} = \eta^{j^2} \eta^{-(k-j)^2} \eta^{k^2}$$

reduces $F_n x$ to one cyclic convolution of length L. Zero-pad $A_j = \eta^{j^2} x_j$ and take the fixed kernel $B_u = \eta^{-u^2}$ on residues $-(n-1) \leq u \leq n-1$, zero elsewhere. There is no aliasing because $L \geq 2n$.

Compute two forward transforms and one inverse, charging the full transform of the fixed kernel as scalar preparation. Chirps are generated successively in $O(n)$ operations, and padding and pointwise products are charged.

The sole supplied root has explicitly specified order

$$D_*(n) = (2n)L\,2^{\lceil \log_2(16L)\rceil} < 64nL^2 < 1024n^3. \tag{10}$$

The chirp, CRT, binary-convolution roots and i are powers of this root; their extraction and local preparation are polylogarithmic. Every scalar denominator is a root power, a nonzero H_j, a nonzero Toeplitz diagonal, a nonzero integer, α_0, or one of the proved nonzero shear coefficients. All integer quantities and addresses fit a fixed number of logarithmic words. These checks retain the original root-supply, coefficient and uniformity conventions.

Substituting the prime estimates gives

$$T(F_n) = O\big(n(\log n)^{1-a}(\log\log n)^{3+a}\big). \tag{11}$$

For $\delta = 63/50000$, $a - \delta = 10^{-6} > 0$. With $t = \log\log n \geq 1$, the positive exponential series gives $e^{(a-\delta)t} \geq (a-\delta)^4 t^4/24$, and $3 + a < 4$. Thus the extra factor in (11) is $O((\log n)^{a-\delta})$. This proves Theorem 1.1. The same program uses a finite direct fallback, with $n = 1$ identity; all choices depend on dimensions and fixed tables, permitting uniform circuit unrolling. No equality $\delta = a$ is asserted.

7. Improvement and source contributions

The saving relative to OpenAI's original all-length theorem is

$$63/50000 - 10^{-13} = 12599999999/10^{13} > 0.$$

OpenAI supplies the exact model, compiler, synchronized-sector and all-length reductions [1, 2]. The construction builds on the public complex588 work and finite provider package [3, 4]. Their contributions include Jacob Sussman's generalized frames and engine; Dugongue's gauges and residual banks; Zhihao Chen's compressed layout; DaysSky's scalar word; Lydakis's bank patterns; Nil Mamano's gauged extension; and Jeremy Kothe and Ryan Shea's integration. The source also credits Dmitri Arkhipov, Douglas Colkitt, Rohan Arun and Swapnil Jain. The bundled Lean source retains these notices.

The formal theorem covers exact complex arithmetic, all positive lengths, unrestricted coefficients and the charged indexing model stated above. The accompanying Lean development and expanded proof give the finite certificate, semantic transfer and all-length argument.

References

1. OpenAI, *An explicit power saving for the exact discrete Fourier transform*, OpenAI Math Release preprint, pinned source, 2026, September 25; immutable source version 1581d4cbdd69a0faf34b2089614fe2a814316a6a.
2. ———, *Finite tensor savings and exact Fourier circuits*, OpenAI Math Release preprint, pinned source, 2026, September 25; immutable source version 1581d4cbdd69a0faf34b2089614fe2a814316a6a.
3. *Finite Network Improvements for the Exact Discrete Fourier Transform*, Public source repository; complex588 result and framework, 2026, Version 9749945f0f3f7fa0edb22065c786f7106953f1db; inspected October 17, 2026.
4. *Big O Speedrun verifier: darkhipo-dft-mixed source package*, Public source repository; entry header, imported goals and C588AT data, 2026, Version 67f090e9dbd158475176b28e06ac26c4a35c38bd; inspected October 17, 2026.

Mathematical supplement

The proof supplements, exact certificate data and arithmetic programs referenced by filename are included in the attached mathematical supplement (ZIP). Open the attachment using a PDF reader with file-attachment support.